

Natural Language Processing (NLP) Lab Manual

Experiment 1: Text Preprocessing (Lowercasing, Stopword Removal, Tokenization)

Aim: Perform the experiment using Python in Jupyter Notebook.

Procedure: 1. Write the theory (5-8 lines). 2. Write the algorithm/steps. 3. Implement the Python code with comments. 4. Execute and capture output screenshot. 5. Write conclusion.

Submission: - Submit printed notebook file. - Attach .ipynb soft copy.

Experiment 2: Word and Sentence Tokenization using NLTK

Aim: Perform the experiment using Python in Jupyter Notebook.

Procedure: 1. Write the theory (5-8 lines). 2. Write the algorithm/steps. 3. Implement the Python code with comments. 4. Execute and capture output screenshot. 5. Write conclusion.

Submission: - Submit printed notebook file. - Attach .ipynb soft copy.

Experiment 3: Stemming and Lemmatization Comparison

Aim: Perform the experiment using Python in Jupyter Notebook.

Procedure: 1. Write the theory (5-8 lines). 2. Write the algorithm/steps. 3. Implement the Python code with comments. 4. Execute and capture output screenshot. 5. Write conclusion.

Submission: - Submit printed notebook file. - Attach .ipynb soft copy.

Experiment 4: Part of Speech (POS) Tagging

Aim: Perform the experiment using Python in Jupyter Notebook.

Procedure: 1. Write the theory (5-8 lines). 2. Write the algorithm/steps. 3. Implement the Python code with comments. 4. Execute and capture output screenshot. 5. Write conclusion.

Submission: - Submit printed notebook file. - Attach .ipynb soft copy.

Experiment 5: Named Entity Recognition (NER) using spaCy

Aim: Perform the experiment using Python in Jupyter Notebook.

Procedure: 1. Write the theory (5-8 lines). 2. Write the algorithm/steps. 3. Implement the Python code with comments. 4. Execute and capture output screenshot. 5. Write conclusion.

Submission: - Submit printed notebook file. - Attach .ipynb soft copy.

Experiment 6: Bag of Words (BoW) Model Implementation

Aim: Perform the experiment using Python in Jupyter Notebook.

Procedure: 1. Write the theory (5-8 lines). 2. Write the algorithm/steps. 3. Implement the Python code with comments. 4. Execute and capture output screenshot. 5. Write conclusion.

Submission: - Submit printed notebook file. - Attach .ipynb soft copy.

Experiment 7: TF-IDF Vectorization

Aim: Perform the experiment using Python in Jupyter Notebook.

Procedure: 1. Write the theory (5-8 lines). 2. Write the algorithm/steps. 3. Implement the Python code with comments. 4. Execute and capture output screenshot. 5. Write conclusion.

Submission: - Submit printed notebook file. - Attach .ipynb soft copy.

Experiment 8: Sentiment Analysis using TextBlob/VADER

Aim: Perform the experiment using Python in Jupyter Notebook.

Procedure: 1. Write the theory (5-8 lines). 2. Write the algorithm/steps. 3. Implement the Python code with comments. 4. Execute and capture output screenshot. 5. Write conclusion.

Submission: - Submit printed notebook file. - Attach .ipynb soft copy.

Experiment 9: Spam Detection using Naive Bayes Classifier

Aim: Perform the experiment using Python in Jupyter Notebook.

Procedure: 1. Write the theory (5-8 lines). 2. Write the algorithm/steps. 3. Implement the Python code with comments. 4. Execute and capture output screenshot. 5. Write conclusion.

Submission: - Submit printed notebook file. - Attach .ipynb soft copy.

Experiment 10: Simple Rule-Based NLP Chatbot

Aim: Perform the experiment using Python in Jupyter Notebook.

Procedure: 1. Write the theory (5-8 lines). 2. Write the algorithm/steps. 3. Implement the Python code with comments. 4. Execute and capture output screenshot. 5. Write conclusion.

Submission: - Submit printed notebook file. - Attach .ipynb soft copy.